

Disproof of TLMC Conjecture 00000000591

v2: the limit statement, formalized over an explicit infinite family

October 2, 2026

Abstract

The new-direction conjecture attached to Wilf's conjecture predicts that on numerical semigroups of embedding dimension 3 the Wilf surplus ratio $(e \cdot n(S) - g(S) - e)/g(S)$ tends to 0 as $g(S) \rightarrow \infty$. We disprove this with an explicit infinite family of complete-intersection semigroups $S_k = \langle 6, 2(6k+5), 3(6k+5) \rangle$, $k = 0, 1, 2, \dots$, for which the ratio equals $\frac{1}{2} - \frac{3}{2g_k} \geq \frac{1}{3}$ for every k and converges to $\frac{1}{2}$. The limit statement and its negation are formalized in core Lean 4 (v4.33.1, no Mathlib) with zero axioms and zero `sorry`; the main theorem is audited as `does not depend on any axioms`.

1 The conjecture, quoted verbatim

From `conjectures/00000000591.md`:

“Definition: Wilf's conjecture. Conjecture: $e \cdot n(S) \geq g(S) + e$ ($n(S)$ the number of nongaps, g the Frobenius number). New-direction conjecture: on embedding-dimension-3 semigroups the ratio of the surplus $e \cdot n(S) - g(S) - e$ to g tends to 0, with an explicit convergence rate.”

Convention 1 (Correspondence declaration). S ranges over numerical semigroups; $e = e(S)$ is the embedding dimension; $g = g(S)$ is the *Frobenius number* (the conjecture's notation; g is not a genus); $n = n(S) = \#\{s \in S : 0 \leq s \leq g(S)\}$ is the number of nongaps in $[0, g]$; surplus = $e \cdot n(S) - g(S) - e$. Every semigroup constructed below is of exactly this kind and is certified by exhaustive search, so no object is substituted for the conjecture's own.

2 The explicit infinite family

For $k = 0, 1, 2, \dots$ put $r_k = 6k + 5$, which is coprime to 6, and

$$S_k = \langle 6, 2r_k, 3r_k \rangle = \langle pq, pr, qr \rangle, \quad (p, q, r) = (2, 3, 6k + 5).$$

Since p, q, r are pairwise coprime, no generator lies in the subsemigroup generated by the other two, so S_k has embedding dimension exactly 3; S_k is a complete intersection (iterated gluing of $\langle 2 \rangle$, $\langle 3 \rangle$, $\langle r_k \rangle$), hence symmetric. For this shape the Frobenius number is

$$g_k = 2pqr - pq - pr - qr = 42k + 29,$$

and symmetry gives exactly $n_k = (g_k + 1)/2 = 21k + 15$ nongaps in $[0, g_k]$. The Wilf surplus at $e = 3$ is therefore

$$\text{surplus}_k = 3n_k - g_k - 3 = 21k + 13, \quad \frac{\text{surplus}_k}{g_k} = \frac{g_k - 3}{2g_k} = \frac{1}{2} - \frac{3}{2g_k}.$$

Theorem 2 (Disproof). *For every k , $g_k \geq 29 > 9$ and $\text{surplus}_k/g_k \geq 1/3$; moreover $\text{surplus}_k/g_k \rightarrow 1/2$. Hence there are embedding-dimension-3 numerical semigroups with arbitrarily large g and surplus ratio bounded below by the positive constant $1/3$: the ratio does not tend to 0. Conjecture 00000000591 (new-direction part) is false.*

Proof. $g_k = 42k + 29 \geq 29$ and $\text{surplus}_k/g_k - 1/3 = (3(21k + 13) - (42k + 29))/(3g_k) = (21k + 10)/(3g_k) \geq 0$. The limit is $\lim_k (42k + 26)/(84k + 58) = 1/2$. $\square$

Fact 3. *The fraction $(\text{surplus}_k, g_k)$ is in lowest terms for every k : a common divisor divides $2 \cdot \text{surplus}_k - g_k = 3$, and $\text{surplus}_k = 21k + 13 \equiv 1 \pmod{3}$.*

3 The Lean formalization

The Lean project (lean4/, core Lean v4.33.1, no Mathlib) proves the discretized negation of the vanishing claim. With `third` = $1/3 \in \mathbb{Q}$, `qzero` = $0 \in \mathbb{Q}$, `surplus` k = $21k + 13$ and `ratioDen` k = $42k + 29$:

`theorem main :`

`exist epsilon : Rat, qzero < epsilon /\ forall N : Nat, exist k : Nat, N <= k /\
 epsilon.den * ratioNum k >= epsilon.num.natAbs * ratioDen k`

ε 's numerator is positive, so $\varepsilon.\text{num.natAbs} = \varepsilon.\text{num} = 1$; both denominators are positive (`third.den` = 3, $g_k \geq 29$), so $\varepsilon.\text{den} \cdot \text{surplus}_k \geq \varepsilon.\text{num} \cdot g_k$ is exactly the cross-multiplied form of $\varepsilon \leq \text{surplus}_k/g_k$: for every N the member $k = N$ violates any decay of the ratio towards 0. The witness is $\varepsilon = 1/3$.

Supporting formalized facts, all audited as `does not depend on any axioms` (`Check.lean`, 32 theorems; no `sorry`, no `native_decide`):

- `exact_identity`: $2 \cdot \text{surplus}_k \cdot g_k = (g_k - 3) \cdot g_k$, i.e. $\text{surplus}_k/g_k = (g_k - 3)/(2g_k)$; `frob_ge_nine`: $g_k \geq 9$; `wilf_relation`: $3n_k = g_k + \text{surplus}_k + 3$.
- `zero_lt_third`: $0 < 1/3$ in $\mathbb{Q}$; `third_le_ratio_zero`: $1/3 \leq 13/29$ and `third_le_ratio_one`: $1/3 \leq 34/71$ as rational-number order statements, by definitional reduction of the rational order on ground values.
- Anchoring certificates that the closed formulas describe the actual semigroups (no object substitution): for $S_0 = \langle 6, 10, 15 \rangle$ an exhaustive search certifies 29 is a gap while $30, \dots, 35$ are nongaps (hence 29 is exactly the Frobenius number) and that the nongap count on $[0, 29]$ is exactly 15; likewise $S_1 = \langle 6, 22, 33 \rangle$ with $g = 71$, $n = 36$.

On the shape of the symbolic inequality: core Lean's order on $\mathbb{Q}$ reduces to boolean cross-multiplication of the `num/den` fields, and core `Rat` literals carry `propext/Quot.sound` through their by `decide` coprimality fields; `omega` and `simp` are avoided for the same reason. The main theorem therefore uses a ground-constructed $\varepsilon = 1/3$ and rational 0, and states the symbolic member inequality in the equivalent exact cross-multiplied $\mathbb{N}$ form; the $\mathbb{Q}$ -order form itself is proved on ground members.

4 Numerical verification

`reproduce.py` verifies by exhaustive enumeration (no closed formula is trusted) that for every $k \leq 300$ and spot checks up to $k = 100,000$: $g_k = 42k + 29$ is exactly the Frobenius number of S_k (gap at g_k , the next 6 integers are nongaps), $n_k = 21k + 15$ is the exact nongap count on $[0, g_k]$, $\text{surplus}_k = 21k + 13$, $\gcd(\text{surplus}_k, g_k) = 1$, and $\text{surplus}_k/g_k \geq 1/3$. The prime-triple family $\langle pq, pr, qr \rangle$ satisfies the same closed formulas:

(p, q, r)	S_k	g	n	surplus	surplus/ g
$(2, 3, 5)$	$\langle 6, 10, 15 \rangle$	29	15	13	$13/29 \approx 0.4483$
$(2, 3, 7)$	$\langle 6, 14, 21 \rangle$	43	22	20	$20/43 \approx 0.4651$
$(2, 5, 7)$	$\langle 10, 14, 35 \rangle$	81	41	39	$39/81 \approx 0.4815$
$(3, 5, 7)$	$\langle 15, 21, 35 \rangle$	139	70	68	$68/139 \approx 0.4892$

All ratios are $\geq 1/3$ and the family values converge to $1/2$ (e.g. $k = 10^6$: $21000013/42000029 = 0.49999996$).

5 What remains true

Wilf's inequality $e \cdot n(S) \geq g(S) + e$ holds on every member here (e.g. $3 \cdot 15 = 45 \geq 29 + 3$); only the attached vanishing-rate direction is false. Moreover the pairing bound $n \leq (g + 1)/2$ gives $\text{surplus}/g \leq (g - 3)/(2g) < 1/2$ for every numerical semigroup of embedding dimension 3, so the symmetric family attains the exact asymptotic ceiling: the conjectured limit 0 fails in the strongest possible way.